ZCAS UNIVERSITY

SCHOOL OF COMPUTING AND APPLIED SCIENCES

CIT1142 EMERGING TRENDS IN ICT

End of Semester Final Examination

Thursday 19th June 2025

09:00 – 12:00 HOURS

TIME ALLOWED: **WRITING – THREE (3) HOURS**
READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer the following question.

SECTION A

QUESTION 1.

Read the following then answer the questions that follow.

The development of grid computing is a cutting-edge technology that brings several benefits to many Universities and Research Institutions around the world. Grid computing enables Universities and Research Institutions to manage Information Technology resources in a centralized multi-core architecture, irrespective of their location in the world. It enables them to solve their ever-increasing computing and storage problems. Universities and Research Institutions would undoubtedly enhance the quality of their output while reducing costs by sharing resources through Grid technology.

Several Universities and Research Institutions believe that grid computing has the capacity to improve research work and other University operations, especially among the growing Institutions in Africa. Nevertheless, many African Universities and Research institutions have not yet embraced and adopted the use of Grid Computing Technology.

Required:

a) Discuss the elements of grid computing. **(20 Marks)**

b) Explain how grid computing is operationalized. **(5 Marks)**

c) Mention and explain the grid computing types that Universities in Africa should focus on. **(15 Marks)**

[Total 40 Marks]

SECTION B

Choose three questions from this section.

QUESTION 2.

The Government of Zambia has noticed that the Chilundu border is a high-risk area for illegal immigrants and that without a proper plan in place, the nation would continue losing over K85,000,000, 000 per month. The Permanent Secretary for Foreign Affairs observed that many foreigners come with forged passports and citizens also cross using forged documents.

You happen to work as an ICT specialist at Bonanza Consultancy Ltd, and you have been asked to help the government of Zambia to find a lasting solution to the problem.

Required:

a) What solution can you recommend for the Department of Immigration and how would it work? **(10 Marks)**

b) Describe the technology that is based on elements of evolution, protein folding and protein function. **(10 Marks)**

[Total 20 Marks]

QUESTION 3.

The world is witnessing a rapid evolution of Technology. You happen to work as a Network consultant at Liquid Telecom and have been invited to make a presentation at Kenneth Kaunda Conference Center on emerging trends in information and communication technology.

Required

a) Explain, using examples, the areas where AI can be applied. **(5 Marks)**

b) What are the core features of pervasive computing? **(5 Marks)**

c) Provide five big players of Cloud computing. **(5 Marks)**

d) Explain what the concept of Mobile computing involves. **(5 Marks)**

[Total 20 Marks]

QUESTION 4.

Discuss Emerging Trends in the following areas:

a) Aviation Industry **(5 Marks)**

b) Production Industry **(5 Marks)**

c) Education Industry **(5 Marks)**

d) Health Industry **(5 Marks)**

[Total 20 Marks]

QUESTION 5.

a) Explain the challenges experienced when using ubiquitous computing **(5 Marks)**

b) Discuss "Digital forensics". **(15 Marks)**

[Total 20 Marks]

END OF EXAMINATION

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